

Campus Network Design – Technical Documentation

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Devices & Their Mapping

Devices & Roles:

Device Name	Device Type	Role
RW-MAIN-BRANCH	Router 2811	Inter-VLAN Routing & DHCP (Main Campus)
RW-CAMPUS-BRANCH	Router 2811	Inter-VLAN Routing & DHCP (Remote Campus)
RW-CLOUD	Router 2811	Cloud connectivity via Static Routing
SW-MAIN-BRANCH	Switch 2960	VLAN segmentation & trunking
SW-CAMPUS-BRANCH	Switch 2960	VLAN segmentation & trunking
PCs, Printers & Servers	End Devices	Connected to respective VLANs

Switch Port Mapping & VLAN Assignment:

Switch	Port	VLAN	Connected Device
SW-MAIN-BRANCH	Gi1/0/1	Trunk	Link to RW-MAIN-BRANCH
SW-MAIN-BRANCH	Gi1/0/2	10	MGM-PC-1
SW-MAIN-BRANCH	Gi1/0/3	20	HR-PC-1
SW-MAIN-BRANCH	Gi1/0/4	30	FINANCE-PC-1
SW-MAIN-BRANCH	Gi1/0/5	40	BUSINESS-PC-1
SW-MAIN-BRANCH	Gi1/0/6	50	E&C-PC-1
SW-MAIN-BRANCH	Gi1/0/7	60	ARTS-PC-1
SW-MAIN-BRANCH	Gi1/0/8	70	STDLAB-PC-1
SW-MAIN-BRANCH	Gi1/0/9	80	IT-SERVER1/2
SW-CAMPUS-BRANCH	Gi1/0/1	Trunk	Link to RW-CAMPUS-BRANCH
SW-CAMPUS-BRANCH	Gi1/0/2	100	STUDLAB-PC-1

Switch	Port	VLAN	Connected Device
SW-CAMPUS-BRANCH	Gi1/0/3	90	STAFF-PC-1

VLAN & DHCP Summary

VLAN ID	Department	Subnet	Gateway	DHCP Pool	Devices Connected
10	MGM	192.168.1.0/24	192.168.1.1	pvlan10	1 PC, 1 Printer
20	HR	192.168.2.0/24	192.168.2.1	pvlan20	1 PC, 1 Printer
30	FINANCE	192.168.3.0/24	192.168.3.1	pvlan30	1 PC, 1 Printer
40	BUSINESS	192.168.4.0/24	192.168.4.1	pvlan40	1 PC, 1 Printer
50	E&C	192.168.5.0/24	192.168.5.1	pvlan50	1 PC, 1 Printer
60	ARTS	192.168.6.0/24	192.168.6.1	pvlan60	1 PC, 1 Printer
70	STDLAB	192.168.7.0/24	192.168.7.1	pvlan70	1 PC, 1 Printer
80	IT	192.168.8.0/24	192.168.8.1	pvlan80	1PC, 2 Servers
90	STAFF	192.168.9.0/24	192.168.9.1	pvlan90	1 PC, 1 Printer
100	STUDLAB	192.168.10.0/24	192.168.10.1	pvlan100	1 PC, 1 Printer

Router-on-a-Stick & DHCP Details

ROAS Subinterfaces:

Subinterface (RW-MAIN-BRANCH)	VLAN	IP Address/24
Fa0/0.10	10	192.168.1.1
Fa0/0.20	20	192.168.2.1
Fa0/0.30	30	192.168.3.1
Fa0/0.40	40	192.168.4.1
Fa0/0.50	50	192.168.5.1
Fa0/0.60	60	192.168.6.1
Fa0/0.70	70	192.168.7.1
Fa0/0.80	80	192.168.8.1

Subinterface (RW-CAMPUS-BRANCH)	VLAN	IP Address/24
Fa0/0.90	90	192.168.9.1
Fa0/0.100	100	192.168.10.1

DHCP Pools:

Pool Name	Network	Default Gateway
pvlan10	192.168.1.0/24	192.168.1.1
pvlan20	192.168.2.0/24	192.168.2.1
pvlan30	192.168.3.0/24	192.168.3.1
pvlan40	192.168.4.0/24	192.168.4.1
pvlan50	192.168.5.0/24	192.168.5.1
pvlan60	192.168.6.0/24	192.168.6.1
pvlan70	192.168.7.0/24	192.168.7.1
pvlan80	192.168.8.0/24	192.168.8.1
pvlan90	192.168.9.0/24	192.168.9.1
pvlan100	192.168.10.0/24	192.168.10.1